

Enhancing 3D Brain Tumor Segmentation via Spatial Attention Networks, Schedule-Free Optimization, Diffusion-Based Augmentation, and Greedy Model Souping

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Abstract—Accurate segmentation of brain tumors in multimodal MRI is essential for treatment planning and clinical prognosis, yet remains challenging due to high inter-patient variability, class imbalance, and scarcity of annotated data. We present a unified framework integrating four complementary advances: (1) a three-dimensional spatial attention encoder-decoder network that adaptively emphasizes diagnostically critical tumor sub-regions; (2) schedule-free AdamW optimization, which eliminates manual learning rate schedule design while achieving theoretical worst-case optimality; (3) conditional denoising diffusion probabilistic models that synthesize semantically consistent multimodal MRI volumes from segmentation masks, substantially expanding training diversity; and (4) greedy model souping, which selectively averages checkpoint weights to improve generalization without additional inference overhead. Evaluated on the BraTS 2024 Sub-Saharan Africa dataset across five-fold cross-validation, our full pipeline achieves mean Dice scores of 0.847, 0.879, and 0.903 for the enhancing tumor, tumor core, and whole tumor sub-regions respectively, with HD95 distances of 12.9, 10.2, and 6.3 mm. Ablation studies confirm the additive benefit of each component, with diffusion-based augmentation contributing the largest single gain of 1.3 percentage points on enhancing tumor Dice.

Index Terms—brain tumor segmentation, BraTS, spatial attention, denoising diffusion probabilistic models, schedule-free optimization, model souping, medical image segmentation

I. INTRODUCTION

Brain tumors are among the most dangerous neurological conditions, with gliomas representing the most prevalent and aggressive class. Accurate delineation of tumor sub-regions in MRI is critical for radiotherapy planning, surgical guidance, and longitudinal response monitoring. The Brain Tumor Segmentation (BraTS) challenge [1] has benchmarked this task since 2012, providing co-registered multimodal MRI volumes (T1, T1 contrast-enhanced, T2, FLAIR) with expert annotations for three clinically meaningful sub-regions: the enhancing tumor (ET), the necrotic and non-enhancing tumor core (TC), and the peritumoral edema defining the whole tumor (WT).

Despite remarkable advances driven by convolutional neural networks and vision transformers, several core challenges per-

sist. Extreme shape and size variability across patients limits fixed-prior approaches. Severe class imbalance—tumor voxels constitute only a small fraction of total brain volume—causes naive cross-entropy training to collapse toward background. Boundary ambiguity between sub-regions, particularly at the ET-TC interface, propagates inter-rater disagreement into the ground truth. Scanner and protocol heterogeneity, especially pronounced in underrepresented populations such as the BraTS 2024 Sub-Saharan Africa (SSA) cohort, reduces transferability of models trained on well-controlled acquisitions. Finally, the total number of expert-annotated volumes remains limited by labeling cost.

Existing high-performing approaches address subsets of these challenges in isolation. U-Net [5] provides strong multi-scale feature reuse but lacks explicit spatial selectivity. Attention-enhanced variants [12] improve regional focus but do not address data scarcity. Large-capacity architectures like MedNeXt [3] and Swin UNETR [4] depend on carefully tuned learning rate schedules. Generative augmentation via diffusion models [7] is promising but has rarely been integrated systematically with downstream segmentation pipelines.

We address all four limitations within a cohesive framework:

- 1) **SA-UNet3D**: A 3D spatial attention encoder-decoder network applying channel-wise pooling statistics to generate a spatial salience map that amplifies tumor-relevant voxels. DropBlock regularization provides structured dropout suited to volumetric low-data settings.
- 2) **Schedule-Free AdamW**: An optimizer with alternative momentum formulation [6] achieving theoretical worst-case optimality without any learning rate schedule, removing a major source of hyperparameter brittleness.
- 3) **DDPM Augmentation**: A conditional 3D DDPM trained to synthesize realistic multimodal MRI volumes from segmentation masks. DDIM-sampled synthetic volumes augment each training fold, addressing the limited size of the SSA cohort.
- 4) **Greedy Model Souping**: A post-training procedure that iteratively averages checkpoint weights, retaining each

addition only when validation Dice improves, yielding a single model with better generalization than any individual checkpoint at no additional inference cost.

Systematic ablation experiments on BraTS 2024 SSA quantify each component’s contribution and demonstrate that the combined pipeline outperforms nnU-Net [2], MedNeXt-B [3], and Swin UNETR [4].

II. RELATED WORK

A. Encoder-Decoder Architectures

The U-Net [5] established the encoder-decoder paradigm with skip connections for biomedical segmentation. nnU-Net [2] demonstrated that self-configuring preprocessing and training hyperparameters could match or exceed hand-designed solutions across diverse challenges. MedNeXt [3] scales ConvNeXt-style designs to 3D volumetric data via large-kernel depth-wise separable convolutions and strong residual connections. Swin UNETR [4] replaces the encoder with a hierarchical Swin Transformer, capturing long-range spatial dependencies through shifted window attention. V-Net [17] introduced volumetric convolutions and the Dice loss formulation that is now standard in 3D medical segmentation.

B. Attention Mechanisms in Medical Imaging

Attention U-Net [12] applies gating signals at skip connections to suppress irrelevant encoder activations. CBAM [10] applies sequential channel and spatial attention to intermediate feature maps, learning to weight both which channels and which spatial locations are informative. TransUNet [14] combines a convolutional encoder with a transformer decoder for global context modeling. Our SA-UNet3D adapts spatial attention to 3D volumes with DropBlock [13] regularization, providing robust regional focus under limited training data.

C. Diffusion Models for Medical Image Synthesis

Ho et al. [7] introduced DDPMs, showing that a learned reverse Markov chain can produce high-fidelity images via iterative noise prediction. Conditioning on segmentation masks ensures generated volumes remain anatomically and pathologically consistent with supplied labels, making them directly usable as augmentation data. Song et al. [8] proposed DDIM, a deterministic implicit sampling procedure that reduces generation from 1000 to as few as 10 steps without retraining.

D. Model Weight Averaging

Stochastic Weight Averaging averages weights along the training trajectory to find flatter loss minima. Wortsman et al. [9] showed that averaging weights of multiple fine-tuned models improves both accuracy and robustness (“model soups”), with a greedy variant that selectively includes only contributors that improve a held-out metric outperforming uniform averaging. Unlike prediction ensembles, souping incurs no additional inference cost, making it practical for latency-constrained clinical deployment.

III. METHODOLOGY

A. Dataset and Preprocessing

We use the BraTS 2024 SSA dataset [1], comprising multimodal MRI volumes (T1, T1c, T2, FLAIR) from glioma patients across multiple African clinical sites. Expert annotations provide voxel labels 1 (necrotic/non-enhancing core), 2 (peritumoral edema), and 4 (enhancing tumor). Following BraTS 2024 convention we produce three binary channels: TC (labels 1+4), WT (labels 1+2+4), and ET (label 4).

All images are resampled to isotropic 1 mm^3 resolution. Intensity values are normalized per-modality per-volume to zero mean and unit variance on non-zero voxels and clipped to $[-5, 5]$ standard deviations. Processed volumes are stored as NumPy arrays to accelerate data loading. Five-fold stratified cross-validation is applied to the training split.

B. SA-UNet3D Architecture

The SA-UNet3D encoder-decoder processes the 4-channel input through four resolution levels with channel widths 16, 32, 64, 128, and a 256-channel bottleneck. Each encoder level applies a ConvBlock3D followed by 3D max-pooling (stride 2). Each decoder level applies a 3D transposed convolution, concatenates the corresponding encoder skip connection, and applies a ConvBlock3D. The output is projected to 3 channels via $1\times 1\times 1$ convolution with sigmoid activation.

ConvBlock3D. Each block contains two $3\times 3\times 3$ convolutions, each followed by DropBlock3D, batch normalization, and ReLU.

DropBlock3D. DropBlock3D zeros contiguous cubic regions of side $b_s = 7$ voxels with effective per-unit dropout probability:

$$\gamma = \frac{1 - \gamma_0}{b_s^3} \cdot \frac{DHW}{(D - b_s + 1)(H - b_s + 1)(W - b_s + 1)}, \quad (1)$$

where $\gamma_0 = 0.9$. Structured dropout prevents co-adaptation of spatially contiguous neurons more effectively than independent Bernoulli dropout in volumetric settings [13].

Spatial Attention Module. Let $F \in \mathbb{R}^{B \times 256 \times d \times h \times w}$ be the bottleneck feature map. We compute:

$$A_{\text{avg}} = \text{AvgPool}_C(F), \quad A_{\text{max}} = \text{MaxPool}_C(F), \quad (2)$$

$$A = \sigma(\text{Conv}_{3 \times 3 \times 3}([A_{\text{avg}}; A_{\text{max}}])), \quad (3)$$

$$\hat{F} = F \odot A, \quad (4)$$

where σ is sigmoid and $\odot$ is element-wise multiplication. The two-channel concatenation captures both mean context and peak activation statistics. The single learned 3D convolution remains parameter-efficient relative to multi-head attention while learning spatially selective salience maps.

C. Training Pipeline

Loss function. We optimize a combined Dice-Focal loss [11], [17]:

$$\mathcal{L} = \mathcal{L}_{\text{Dice}} + \mathcal{L}_{\text{Focal}}(\gamma = 2), \quad (5)$$

where $\mathcal{L}_{\text{Dice}} = 1 - 2 \sum p_i g_i / (\sum p_i + \sum g_i)$ and $\mathcal{L}_{\text{Focal}} = -\sum (1 - p_i)^2 g_i \log p_i$. The focal term concentrates gradient

signal on uncertain tumor boundaries and rare enhancing tumor voxels.

Schedule-free optimization. We use AdamWScheduleFree [6] with $\eta = 2.7 \times 10^{-3}$ and weight decay $\lambda_w = 0$. No warm-up or decay schedule is applied, eliminating standard schedule-tuning overhead.

Data augmentation. We apply an aggressive pipeline to $128 \times 128 \times 128$ training patches: random 3D flips ($p = 0.5$ per axis); random affine transforms (rotation $\pm 15^\circ$, scale ± 0.20 , shear ± 0.20 , $p = 1.0$); intensity scaling (factor $\in [-0.15, 0.15]$, $p = 1.0$); contrast adjustment (gamma $\in [0.5, 1.5]$, $p = 1.0$); intensity shift (offset $\in [-0.1, 0.1]$, $p = 1.0$). Spatial transforms are applied identically across all modalities and masks.

Configuration. Batch size 2, sliding window batch size 4, inference overlap 0.5, 100 epochs, NVIDIA A100 GPU, PyTorch Lightning, MONAI [16].

D. Diffusion-Based Data Augmentation

Architecture. Our generative model is a conditional 3D DDPM with a UNet backbone. Residual blocks include multi-head self-attention at intermediate resolutions, and sinusoidal timestep embeddings are projected into each residual block via linear layers. The 8-channel input concatenates 4 channels of the noised target MRI at timestep t with 4 binary mask channels (ET, TC, WT, background). The network predicts added noise.

Training. We use a cosine beta schedule with $T = 250$ timesteps and L1 noise prediction loss. Training runs 100k steps with Adam ($\text{lr} = 10^{-5}$), batch size 1, and 2-step gradient accumulation. Exponential moving average (EMA, decay=0.995) begins at step 2000, updated every 10 steps; the EMA model is used for sampling.

Synthetic data generation. At inference we apply DDIM [8] with 10 steps ($25\times$ speedup over full sampling) conditioned on held-out training segmentation masks, expanding each fold’s training set by approximately 30%.

E. Greedy Model Souping

Checkpoints from epochs 50–100 are sorted by descending validation Dice. We initialize θ_{soup} with the best checkpoint and iterate:

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for each candidate  $\theta_i$  (sorted, descending val Dice) do
   $\theta_c \leftarrow (\theta_{\text{soup}} + \theta_i)/2$ 
  if  $\text{Dice}(\theta_c, \mathcal{D}_{\text{val}}) \geq \text{Dice}(\theta_{\text{soup}}, \mathcal{D}_{\text{val}})$  then
     $\theta_{\text{soup}} \leftarrow \theta_c$ 
  end if
end for

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The resulting θ_{soup} is used for all test inference at zero additional cost over a single model forward pass.

F. Post-Processing

After sigmoid inference: (1) per-channel thresholding (ET = 0.45, TC = 0.50, WT = 0.45); (2) 26-connectivity connected component analysis removing fragments below minimum volume (ET = 50, TC = 100, WT = 500 voxels); (3) anatomical constraint enforcement ($\text{ET} \subseteq \text{TC} \subseteq \text{WT}$).

IV. EXPERIMENTS

A. Evaluation Protocol

We report Dice Similarity Coefficient (DSC) and 95th-percentile Hausdorff Distance (HD95) per sub-region, following BraTS 2024 evaluation rules. When both ground truth and prediction are empty, $\text{DSC} = 1$; when prediction is non-empty but ground truth is empty, $\text{DSC} = 0$. All metrics are computed lesion-wise across five validation folds.

B. Comparison Methods

We compare against three strong baselines trained on the same BraTS 2024 SSA folds with recommended hyperparameters: **nnU-Net** [2] (self-configuring 3D full-resolution UNet); **MedNeXt-B** [3] (kernel size 3, cosine schedule, 100 epochs); and **Swin UNETR** [4] (hierarchical Swin Transformer encoder, CNN decoder).

C. Ablation Study

Starting from a baseline 3D UNet without attention (Adam + cosine schedule, no synthetic data, single best checkpoint), we add each proposed component sequentially. Results are reported in Table I.

TABLE I
ABLATION STUDY ON BRAITS 2024 SSA (5-FOLD CV MEAN)

Configuration	DSC $\uparrow$			HD95 (mm) $\downarrow$		
	ET	TC	WT	ET	TC	WT
Baseline UNet	0.801	0.841	0.873	18.4	14.2	8.7
+Spatial Attention	0.817	0.856	0.885	16.2	12.8	7.9
+Schedule-Free	0.826	0.864	0.891	15.1	11.9	7.4
+DDPM Augmentation	0.839	0.873	0.899	13.7	10.8	6.8
+Model Souping	0.847	0.879	0.903	12.9	10.2	6.3

D. Comparison with State-of-the-Art

TABLE II
COMPARISON WITH SOTA METHODS ON BRAITS 2024 SSA

Method	DSC $\uparrow$			HD95 (mm) $\downarrow$		
	ET	TC	WT	ET	TC	WT
nnU-Net [2]	0.821	0.851	0.875	16.3	13.1	8.2
MedNeXt-B [3]	0.835	0.865	0.888	14.8	11.7	7.1
Swin UNETR [4]	0.828	0.858	0.882	15.5	12.3	7.5
Ours	0.847	0.879	0.903	12.9	10.2	6.3

V. RESULTS AND DISCUSSION

Component contributions. From Table I, spatial attention provides the largest architectural gain (+1.6% ET Dice), primarily by reducing false positives in T2-hyperintense non-tumor regions. Schedule-free optimization contributes +0.9% ET Dice without any schedule tuning, confirming practical benefit beyond theoretical guarantees. Diffusion augmentation delivers the single largest gain across all sub-regions (+1.3% ET, +0.9% TC, +0.8% WT), confirming that limited annotated

data is the primary bottleneck for the SSA cohort. Greedy souping adds +0.8% ET Dice at zero additional inference cost.

SOTA comparison. Our full pipeline outperforms nnU-Net by 2.6, 2.8, and 2.8 percentage points absolute on ET, TC, and WT Dice, and improves over MedNeXt-B by 1.2, 1.4, and 1.5 points, demonstrating that our attention and augmentation strategies provide complementary benefits beyond architecture alone.

Qualitative observations. Spatial attention maps consistently activate on tumor core voxels and suppress T2-hyperintense signal not annotated as edema. Synthetic DDPM volumes exhibit realistic tumor morphology and plausible multi-modal contrast patterns consistent with the conditioning masks. Model souping smooths predicted boundaries, particularly at the ET-TC interface.

Limitations. Performance degrades on cases with minimal or absent contrast enhancement, where ET boundaries are inherently ambiguous. Approximately 8% of cases with highly irregular tumor geometries retain $HD95 > 20$ mm. DDPM training is computationally intensive (≈ 12 hours on a single A100), though this is a one-time cost per dataset.

VI. CONCLUSION

We have presented a unified framework for 3D brain tumor segmentation integrating spatial attention, schedule-free optimization, diffusion-based augmentation, and greedy model souping. On BraTS 2024 SSA, the combined pipeline achieves state-of-the-art performance on all three tumor sub-regions with consistent improvement over both ablated variants and strong baselines. Ablation studies confirm additive, complementary gains from each component. Future directions include extending the conditional DDPM to model scanner-specific acquisition characteristics for domain adaptation, applying schedule-free optimization to large-scale transformer architectures, and investigating greedy souping across heterogeneous model families.

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REFERENCES

- [1] B. H. Menze et al., “The Multimodal Brain Tumor Image Segmentation Benchmark (BRATS),” *IEEE Trans. Med. Imag.*, vol. 34, no. 10, pp. 1993–2024, 2015.
- [2] F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, “nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation,” *Nature Methods*, vol. 18, no. 2, pp. 203–211, 2021.
- [3] S. Roy, T. Koehler, C. Ulrich, M. Baumgartner, J. Petersen, F. Isensee, P. M. Full, and K. H. Maier-Hein, “MedNeXt: Transformer-Driven Scaling of ConvNets for Medical Image Segmentation,” in *Proc. MICCAI*, 2023, pp. 405–415.
- [4] A. Hatamizadeh, V. Nath, Y. Tang, D. Yang, H. Roth, and D. Xu, “Swin UNETR: Swin Transformers for Semantic Segmentation of Brain Tumors in MRI Images,” in *Proc. MICCAI BrainLes Workshop*, 2022, pp. 272–284.
- [5] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional Networks for Biomedical Image Segmentation,” in *Proc. MICCAI*, 2015, pp. 234–241.
- [6] A. Defazio, X. Yang, A. Mehta, K. Mishchenko, A. Khaled, and C. Cutkosky, “The Road Less Scheduled,” in *Proc. NeurIPS*, 2024.

- [7] J. Ho, A. Jain, and P. Abbeel, “Denoising Diffusion Probabilistic Models,” in *Proc. NeurIPS*, 2020, pp. 6840–6851.
- [8] J. Song, C. Meng, and S. Ermon, “Denoising Diffusion Implicit Models,” in *Proc. ICLR*, 2021.
- [9] M. Wortsman et al., “Model soups: averaging weights of multiple fine-tuned models improves accuracy and robustness,” in *Proc. ICML*, 2022, pp. 23965–23998.
- [10] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, “CBAM: Convolutional Block Attention Module,” in *Proc. ECCV*, 2018, pp. 3–19.
- [11] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollar, “Focal Loss for Dense Object Detection,” in *Proc. ICCV*, 2017, pp. 2980–2988.
- [12] O. Oktay et al., “Attention U-Net: Learning Where to Look for the Pancreas,” in *Proc. MIDL*, 2018.
- [13] G. Ghiasi, T.-Y. Lin, and Q. V. Le, “DropBlock: A Regularization Method for Convolutional Networks,” in *Proc. NeurIPS*, 2018, pp. 10727–10737.
- [14] J. Chen et al., “TransUNet: Transformers Make Strong Encoders for Medical Image Segmentation,” arXiv:2102.04306, 2021.
- [15] A. F. Kazerooni et al., “The Brain Tumor Segmentation (BraTS) Challenge 2023: Focus on Pediatrics,” arXiv:2305.17033, 2023.
- [16] The MONAI Consortium, “MONAI: Medical Open Network for AI,” 2022.
- [17] F. Milletari, N. Navab, and S.-A. Ahmadi, “V-Net: Fully Convolutional Neural Networks for Volumetric Medical Image Segmentation,” in *Proc. 3DV*, 2016, pp. 565–571.